A-A-53905A
7 September 1990
SUPERSEDING
A-A-53905
19 May 1989

COMMERCIAL ITEM DESCRIPTION

ELECTROCARDIOGRAPH, AUTOMATIC

The General Services Administration has authorized the use of this commercial item description.

This Commercial Item Description covers an electrocardiograph suitable to diagnose cardiac abnormalities.

Salient Characteristics:

The electrocardiograph shall be designed to diagnose cardiac abnormalities, document patients' response to treatment and alert the medical staff when trends or changes in heart functions occur. Shall be a solid state, dual speed, direct writing, portable, AC/DC powered electrocardiograph with 5, 10, and 20 mm/mV sensitivity selections, and at least 5 or 10 mm/mV in the automatic mode. Sensitivity may be 5 or 10 mm/mV for chest leads, both modes. Shall operate on a nominal 115 VAC, 60 Hz and an internal, rechargeable DC power source. The battery shall recharge while in the unit. When used on battery power, the unit shall operate for a minimum of 45 minutes. The electrocardiograph shall meet the requirements of UL544, and comply with AHA or AAMI specifications for instruments used in electrocardiography. Shall be able to record leads I, II, III, aVR, aVL, aVF, and V_1 through V_6. Shall have automatic lead markings and automatic calibration pulses of 1 mV amplitude. Shall have selectable paper speeds of 25 and 50 mm/sec plus or minus 5 percent. Shall include a control to permit manual selection of leads.

The electrocardiograph shall be provided with a functional battery pack, an AC artifact eliminator and a non-cue timer. The electrocardiograph shall be protected against

AMSC N/A FSC 6515

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defibrillator countershock up to 400 Joules. The electrocardiograph shall contain a recording device that uses 5 cm grid paper (see note), and a control to permit manual selection of each lead (I, II, III, aVR, aVL, and V_1 through V_6).

The electrocardiograph shall filter 50/60 Hz interference from input line power.

Shall be equipped with thermal array dot printer or stylus to produce tracings. If a stylus type recorder is used, shall include controls for stylus heat and position and include a suitable adjustment for stylus pressure. The stylus shall be operator replaceable and include replacement adjustment tools.

The electrocardiograph shall be supplied with recording paper compatible with the printer provided:

For heat stylus printers, the chart paper shall be 63 mm wide, 150 ft. long, with a 50 mm grid width ruled 1 mm by 1 mm square with 5 mm increments enhanced (see notes).

For thermal array (dot matrix printers), the recording chart paper shall be at least 50 mm wide, 150 ft. long, with at least 40 mm grid width ruled in 1 mm by 1 mm square with 5 mm increments enhanced.

Accessories. The electrocardiograph shall be provided with the following accessory items:

One ten foot, three conductor line cord, fitted with hospital grade (green dot), three blade "U" ground male plug.

One ten lead, patient cable.

One set of reusable electrodes (four limb and six cup).

One set of four limb electrode straps.

Two tubes of electrode paste.

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One replacement power fuse (if required).

One minimum packaging configuration of disposable pre-gelled electrodes.

One carrying case.

One spare battery of each type used.

One unit of issue of recording paper.

One ECG ruler.

Extra stylus (if applicable).

The electrocardiograph shall operate within the following parameters:

Liniarity may not exceed one percent.

The frequency response shall be 3db down from 0.05 to 100 Hz (in accordance with AHA standards).

The decay time shall be such that at least 3 mm shall remain of a 10 mm deflection after an elapsed time of 3.2 seconds. Decay shall not exceed 5 percent in the first 0.2 seconds.

The minimum common mode rejection ratio shall be 100 dB with respect to chassis ground. Measurement to be 60 Hz with 5K OHM of misbalance in lead source resistance.

Manuals:

The unit shall be provided with two (2) complete sets of operator's, maintenance and training manuals (if applicable) which will contain essentially the same data used by the manufacturer's or dealer's field service engineers/technicians to install, operate, service and repair the end item.

The operator's and maintenance manuals shall include installation instructions, set up procedures, operating instructions, all operator maintenance requirements, lists

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of available accessories (including part number and item identification), maintenance and calibration procedures, full schematics, wiring diagrams and internal part identification. All instructions and data will be in the English language and all schematics and wiring diagrams will use American electrical and electronic symbols.

The style and format of the manuals shall be that of the supplier's standard commercial practice; however, any required information not normally included in the supplier's standard manual, may be provided as attachments to the standard manual.

Repair Parts:

The unit shall be provided with a list of recommended spare/repair parts identified as high mortality items. The parts list provided should include all items and quantities required to support the maintenance requirements, per unit, in a fully operational (maximum utilization) state for a period of one (1) year. The source of supply will also be identified.

The unit shall be provided with a list of all consumable items including part/product number and source necessary to insure full system operation, at maximum usage level, for 16 hours per day for a period of 30 days. This list shall include all consumable items that can be expected to be used, regardless of source of supply or manufacturer.

Workmanship. The electrocardiograph shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One electrocardiograph, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of S-1. The unit of product for examination purposes shall be one electrocardiograph.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of S-1. The unit of product for test purposes shall be one electrocardiograph.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

Performance tests. The unit shall be set up and tested using a heart signal simulator. Tracings shall be made at each of the sensitivities and both speeds which are specified. Speeds and sensitivities shall be indicated on the tracings and the tracings shall be forwarded to the Contracting Officer for examination and approval. The tracings shall be clear, dark, and sharp and shall be free of AC interference, wandering, tremor and burns.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented be Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers

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and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTES: 1. The following National Stock Number is covered by this document.

6515-01-115-0025

2. The electrocardiograph is compatible with the standardized electrocardiograph recording paper, NSN: 6515-00-584-3076.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project Number: 6515-4863

Location: ECGCID.17D

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53905A	ELECTROCARDIOGRAPH, AUTOMATIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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